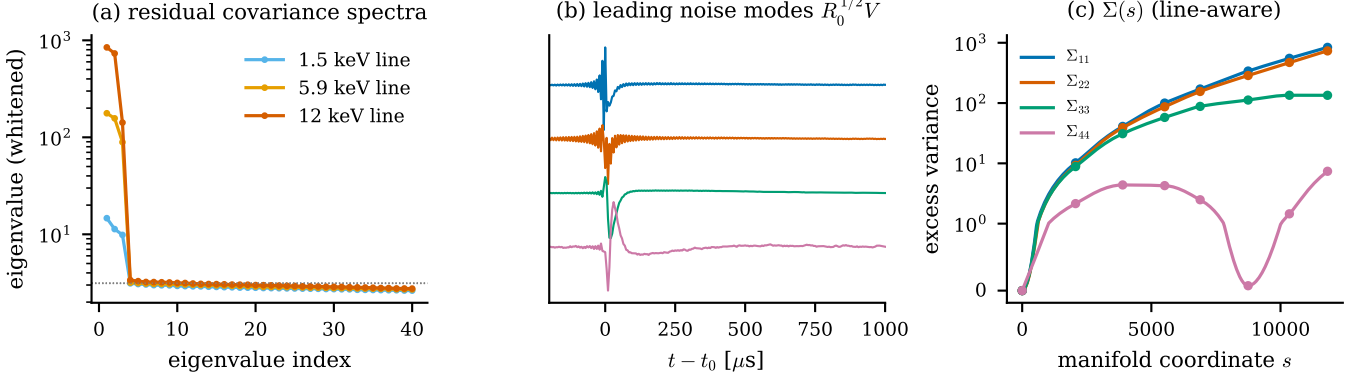
(a) residual covariance spectra
1.5 keV line
5.9 keV line
12 keV line
eigenvalue (whitened)
eigenvalue index
(b) leading noise modes $R_0^{1/2}V$
$t - t_0$ [μs]
(c) $\Sigma(s)$ (line-aware)
Σ_{11}
Σ_{22}
Σ_{33}
Σ_{44}
excess variance
manifold coordinate s